THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 15	Squares from 15^2 to 29^2
2	**(a)** 2^8	$2 \times 2^7 = 2^8$
3	**(d)** 5	$3b + 4 = 19$
4	**(b)** 90°	One part $= 18^\circ$
5	**(a)** 41	LCM plus the common remainder
6	**(d)** 62	Next square is $36^2 = 1296$
7	**(b)** 4	$2x + 1 = 9$
8	**(c)** 5	Number of binary digits of the largest weight to be measured
9	**(c)** 60°	$x + 2x = 180$
10	**(a)** 57	71 multiples up to 500 minus 14 multiples below 100
11	**(b)** 77	Previous square is $50^2 = 2500$
12	**(d)** 7	$2^4 \times 5^7 = 10^4 \times 125$
13	**(a)** 32_5	Convert every number to decimal and compare
14	**(c)** 130°	A diagonal bisects the vertex angles; $\angle BCD = 2x$ and $\angle B$ is supplementary to it
15	**(c)** 112	Count multiples from 100 to 999

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Next square 3969 (1); add 69, root 63 (1).	1+1
17	$243 = 3^5$ (1); $2x - 1 = 5$ gives $x = 3$ (1).	1+1
18	$2b + 3 = 13$ (1); $b = 5$ (1).	1+1
19	Opposite angles equal gives $x = 12$ (1); $\angle A = 62^\circ$, so $\angle B = 180 - 62 = 118^\circ$ (1).	1+1
20	LCM(20, 25) $= 100$ (1); number $= 100 + 3 = 103$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Side $= \sqrt{1225} = 35$ m (1); perimeter $= 4 \times 35 = 140$ m (1); cost $= 140 \times 40 =$ ₹5600 (1).	1+1+1
22	Thickness doubles at each fold: 0.1×2^{10} (1); $= 102.4$ mm (1); in cm $= 10.24$ cm (1).	1+1+1
23	$45 = 101101_2$ (1); weights used: 1, 4, 8, 32 g (1); each 1 means the weight is used (1).	1+1+1
24	$\angle B = 130^\circ$ (1); $\angle PAB = 25^\circ$ and $\angle PBA = 65^\circ$ (1); $\angle APB = 180 - (25 + 65) = 90^\circ$ (1).	1+1+1
	OR: 247247 $= 247 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 35321, 22477, 19019) (1).	1+1+1
25	Number $= 6^3 = 216$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
	OR: 2 min $= 120$ s (1); distance $= 3 \times 10^8 \times 120$ (1); $=3.6 \times 10^{10}$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	LCM $= 24 = 2^3 \times 3$ (2); multiplier to make a perfect square 6 (2); required number 144 (1).	2+2+1
	OR: Number $= 5 \times 3^4$ (2); value 405 (2); it is $3^4 = 81$ times the original (1).	2+2+1
27	Digits are all less than b, so $b > 4$ (1); $2b^2 + 3b + 4 = 69$ (2); trying $b = 5$ gives $50 + 15 + 4 = 69$ (1); so $b = 5$ (1).	1+2+1+1
	OR: One part $= 360 \div 10 = 36^\circ$ (1); angles 36°, 72°, 108°, 144° (2); $\angle A + \angle D = 36 + 144 = 180^\circ$ (1); these are co-interior angles, so AB $\parallel$ DC (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 29 and 30	$(n+1)^2 - n^2 = 2n+1$ gives $n = 29$
2	**(c)** 3	Ones digits repeat in the cycle [7, 9, 3, 1]
3	**(b)** 4	77 = 1001101_2
4	**(b)** 50°	$\angle AOB = 180 - x$; OA = OB, so $\angle OAB = x/2$
5	**(b)** 111	Test palindromes 101, 111, 121, ...
6	**(a)** 3	A k-digit square has $\lceil k/2 \rceil$ digits in its root
7	**(d)** 27	$x = 5$, so 3^3
8	**(b)** MCMXCIV	Break 1994 into thousands, hundreds, tens and units
9	**(a)** 110°	$\angle A + \angle D = 180^\circ$ gives x; base angles $\angle C = \angle D$
10	**(b)** 1	Divisible by 10 and 9
11	**(a)** -9	$(-9)^3 = -729$
12	**(b)** $\frac{6}{5}$	$\left(\frac{1}{2} + \frac{1}{3}\right)^{-1}$
13	**(d)** 11000_2	Add 1 and carry in base 2
14	**(a)** 6	Perimeter $= 4\times$ side
15	**(d)** 5	Apply the routine and count

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$(n+1)^2 - n^2 = 2n+1$ (1) so $n = 37$; numbers 37 and 38 (1).	1+1
17	Convert to the same power or usual form: 340000 (1); 3.4×10^5 (1).	1+1
18	$90 = 2\times45+0$; $45 = 2\times22+1$; $22 = 2\times11+0$; $11 = 2\times5+1$; $5 = 2\times2+1$; $2 = 2\times1+0$; $1 = 2\times0+1$ (1); 1011010_2 has 4 ones (1).	1+1
19	One part $= 360 \div 20 = 18^\circ$ (1); largest $= 126^\circ$, which is obtuse (1).	1+1
20	72 multiples up to 800 (1); minus 13 below 150, giving 59 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Decimal values 98 and 57 (1); sum 155 (1); 1110_5 (1).	1+1+1
22	$10x - 60 = 360$ (1); $x = 42$ (1); angles 42°, 74°, 106°, 138° (1).	1+1+1
23	LCM $= 24$ (1); 9999 divided by it leaves remainder 15 (1); number 9984 (1).	1+1+1
24	Side $= \mathrm{HCF}(36, 48) = 12$ m (1); area 1728 m² (1); tiles $= \frac{1728}{144} = 12$ (1).	1+1+1
	OR: Take 2^5 common in the numerator: $2^5(2+1)$ (1); cancel $(2+1)$ (1); $2^5 \div 2^4 = 2$ (1).	1+1+1
25	Each of the 8 places has 2 choices, so $2^8 = 256$ numbers (2); largest 255 (1).	2+1
	OR: Let the smaller be x: $x + 2x = 180$ (1); $x = 60^\circ$ (1); angles 60°, 120°, 60°, 120° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Splitting 45 into coprime factors and applying the rules (2); possibilities: a = 2, b = 0 (number 4230), a = 6, b = 5 (number 4635) (2); verification (1).	2+2+1
	OR: LCM $= 36 = 2^2 \times 3^2$ (2); multiplier to make a perfect cube 6 (2); required number 216 (1).	2+2+1
27	A is heavier (1); ratio $= \frac{6}{3} \times 10^{24-22} = 200$ times (2); total 6.03×10^{24} kg (2).	1+2+2
	OR: Decimal values 187 and 96 (1); difference 91 (2); 331_5 (1); check by converting back (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	7^3 ends in 3
2	**(c)** 0.5×10^6	Convert each to usual form and compare
3	**(c)** 3	The last place has value 3, so the number is a multiple of 3
4	**(b)** isosceles trapezium	Isosceles trapezium has equal diagonals
5	**(c)** 99	9 and 11 have no common factor
6	**(b)** 33	$14 + 19 = 33$
7	**(b)** 40	$2^{x+3} = 2^x \times 2^3$
8	**(c)** 6	$2b + 5 = 17$
9	**(c)** 120°	One part $= 30^\circ$
10	**(b)** 67	LCM plus the common remainder
11	**(b)** 23	Squares from 32^2 to 54^2
12	**(c)** 3^6	$3 \times 3^5 = 3^6$
13	**(b)** 7	Number of binary digits of the largest weight to be measured
14	**(d)** 45°	$x + 3x = 180$
15	**(a)** 59	75 multiples up to 900 minus 16 multiples below 200

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Between n^2 and $(n+3)^2$ there are 44 numbers (1); squares inside: 49 and 64, so non-squares $= 42$ (1).	1+1
17	Ones digits of powers of 2 repeat as [2, 4, 8, 6] (1); 47 leaves position 3 in the cycle, so the digit is 8 (1).	1+1
18	CXXVII = 127 (1); 1111111_2 (1).	1+1
19	$\angle D = \angle B = 100^\circ$ (1); $\angle C = 360 - 70 - 2 \times 100 = 90^\circ$ (1).	1+1
20	Middle number $= 45 \div 5 = 9$ (1); numbers 7, 8, 9, 10, 11 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	LCM $= 24$ (1); multiples from 100 to 999: $41 - 4$ (1); 37 (1).	1+1+1
22	Side $= \sqrt{2304} = 48$ m (1); perimeter $= 4 \times 48 = 192$ m (1); cost $= 192 \times 35 =$ ₹6720 (1).	1+1+1
23	Thickness doubles at each fold: 0.2×2^8 (1); $= 51.2$ mm (1); in cm $= 5.12$ cm (1).	1+1+1
24	$77 = 1001101_2$ (1); weights used: 1, 4, 8, 64 g (1); each 1 means the weight is used (1).	1+1+1
	OR: $\angle B = 110^\circ$ (1); $\angle PAB = 35^\circ$ and $\angle PBA = 55^\circ$ (1); $\angle APB = 180 - (35 + 55) = 90^\circ$ (1).	1+1+1
25	$358358 = 358 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 51194, 32578, 27566) (1).	1+1+1
	OR: Number $= 7^3 = 343$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Opposite sides are equal: $3x + 4 = 2x + 9$ gives $x = 5$ (2); AB = CD $= 19$ cm (1); perimeter $= 2(19 + 12) = 62$ cm (2).	2+1+2
	OR: 832-238=594 (1); reverse of 594 is 495 (1); 594+495=1089 (1); in general the difference is $99(a - c)$, whose digits are $(a - c - 1), 9, (10 - a + c)$ and the sum is always 1089 (2).	1+1+1+2
27	LCM $= 180 = 2^2 \times 3^2 \times 5$ (2); multiplier to make a perfect square 5 (2); required number 900 (1).	2+2+1
	OR: Number $= 2 \times 4^3$ (2); value 128 (2); it is $4^3 = 64$ times the original (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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